Biological Molecules

- Identify the chemical elements present in carbohydrates, proteins and lipids
- Describe the structure and function of starch, cellulose and glycogen as large molecules made up from glucose
- Describe the structure of proteins and lipids as large molecules made up from smaller basic units: protein from amino acids, and lipid from fatty acids and glycerol

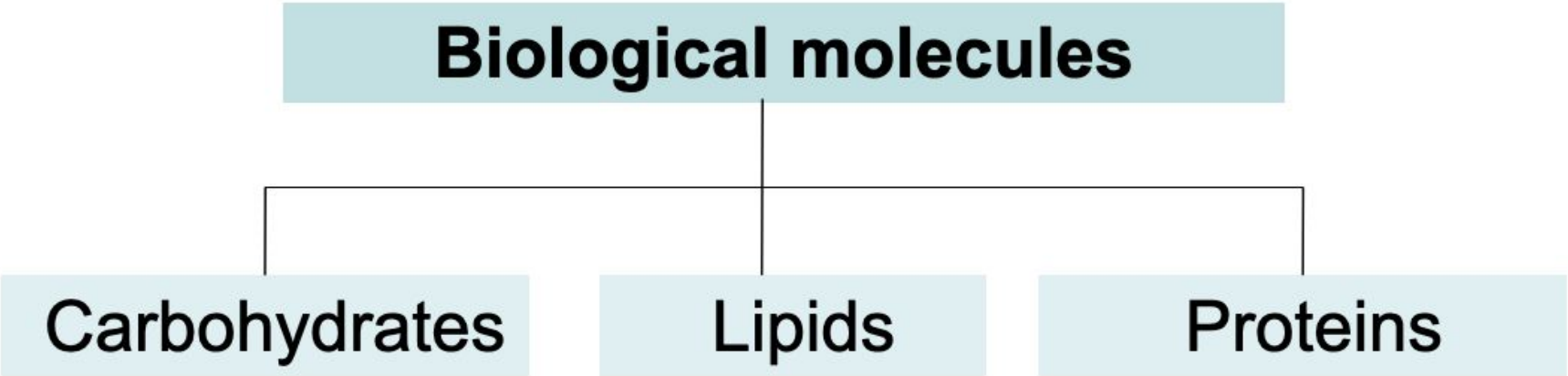
Starter: What’s on my plate?

“What do these foods give us?”

“What molecules do you think make up these foods?”

What are biological molecules?

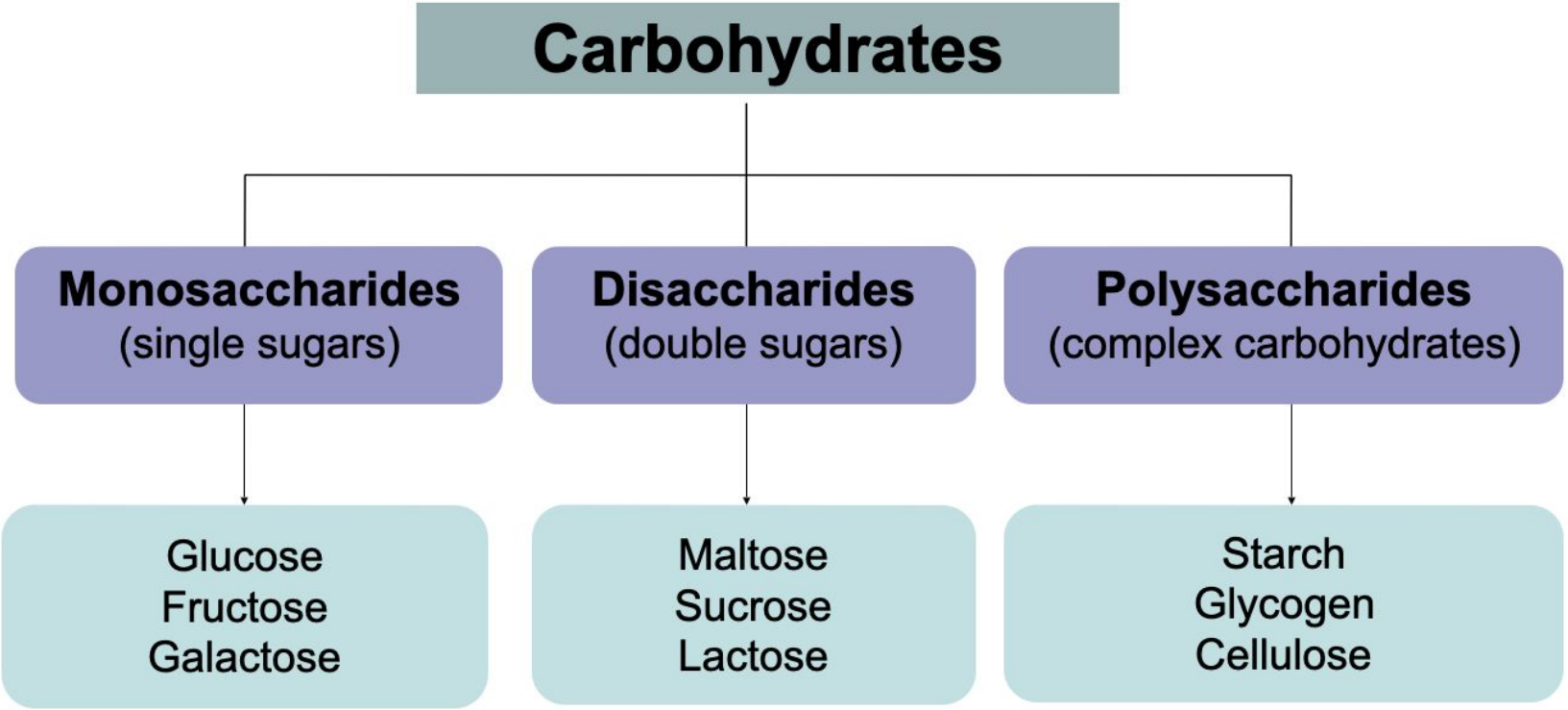
Biological molecules are organic molecules essential for life, such as proteins, lipids, carbohydrates, and nucleic acids. They are crucial for cellular structure and function, providing energy, storing genetic information, and carrying out metabolic processes.


Biological molecules
Carbohydrates
Lipids
Proteins

Carbohydrates

Carbohydrates

- Carbohydrates are organic molecules made up of:
  - **carbon**
  - **hydrogen**
  - **oxygen**
- The general formula for carbohydrates is $\mathbf{C_nH_{2m}O_m}$.
- The ratio of hydrogen atoms to the oxygen atoms in carbohydrates is **2:1**.



Extension: All monosaccharides and disaccharides (except sucrose) are **reducing sugars**.

Types of complex carbohydrates

Starch

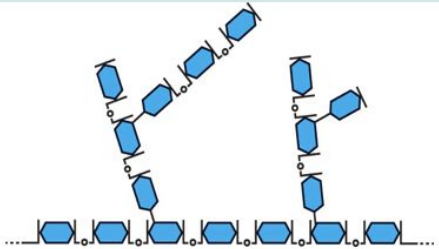
- A storage form of carbohydrates in plants
- Found in storage organs in plants

Glycogen

- A storage form of carbohydrates in mammals
- Digested to form glucose when needed

Cellulose

- Cellulose cell wall protects plant cells
- Serve as dietary fibre to prevent constipation



Functions of carbohydrates

Carbohydrates are needed:

- to provide **energy** for cell activities
- to form **supporting structures** (e.g. cell wall)*
- for conversion into other **organic compounds** (e.g. amino acids and lipids)
- to form **nucleic acids** (e.g. DNA)
- to synthesise **lubricants**
- to synthesise **nectar** in flowers*

*(not required)

Lipids

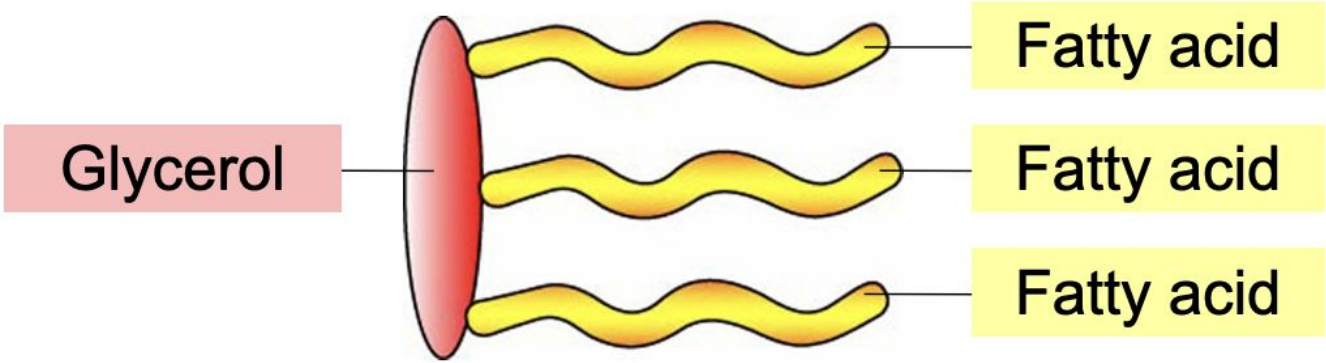
Lipids

- Lipids are organic compounds made up of:
  - **carbon**
  - **hydrogen**
  - **oxygen**
- Lipids contain much **less** oxygen in proportion to hydrogen.
- The proportions of the elements in fats are **not fixed**.

What are lipids made up of?

A fat molecule is made up of:

- **glycerol**
- **fatty acid chains**



Functions of lipids

- a source and **store of energy**
- act as an **insulating material** to prevent excessive heat loss
- are a **solvent** for fat soluble vitamins and some hormones
- form the main part of **cell membranes**
- help **reduce water loss** from skin surface

Proteins

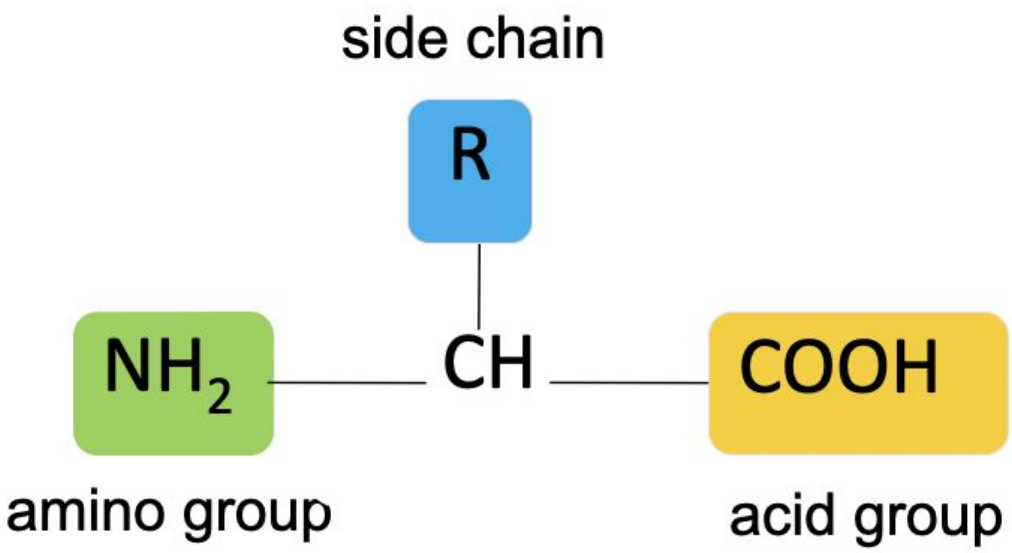
- Proteins are organic compounds made up of:
  - **carbon**
  - **hydrogen**
  - **oxygen**
  - **nitrogen**
- Each protein molecule is made up of smaller basic units called **amino acids**.

Proteins

What are proteins?

Amino acids, the building blocks of proteins, are made up of:

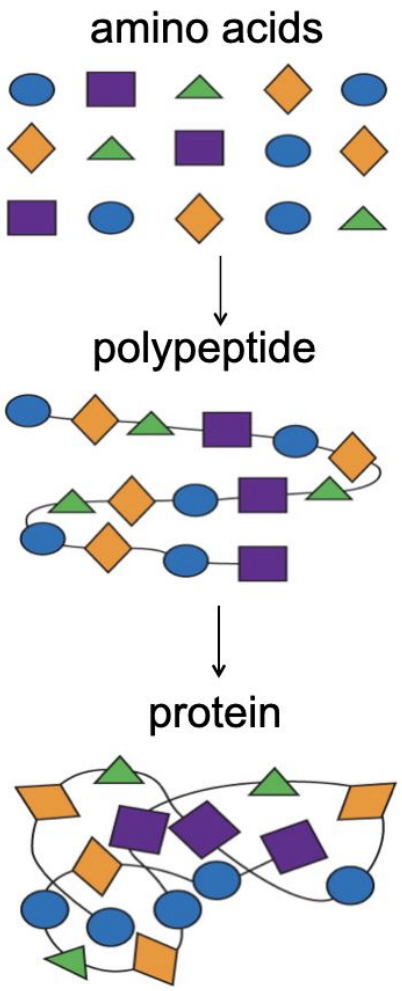
- **an amino group**
- **an acid group**
- **a side chain**



- **Amino acids** are the building blocks of proteins

- Peptide bonds link up amino acids to form a **polypeptide**

- Two or more polypeptide chains fold to form a **3-D molecule**, called **protein**



Functions of proteins

- the synthesis of new protoplasm for growth and repair of worn-out cells
- the synthesis of enzymes and some hormones
- the synthesis of antibodies

Food Tests

- Identify the presence of glucose, starch, protein and fat

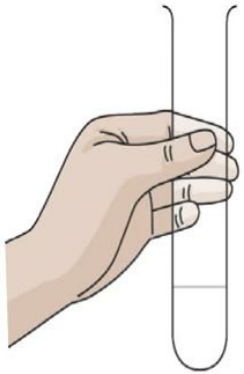
How can we test for carbohydrates?

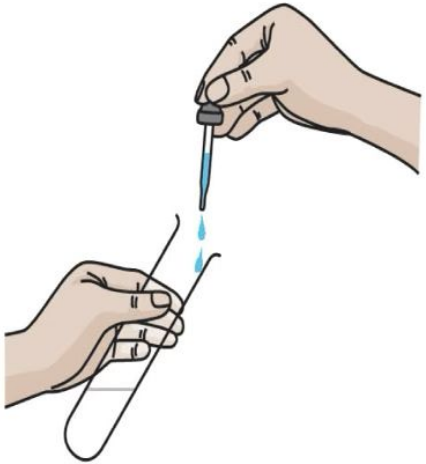
By carrying out the **Benedict's test**

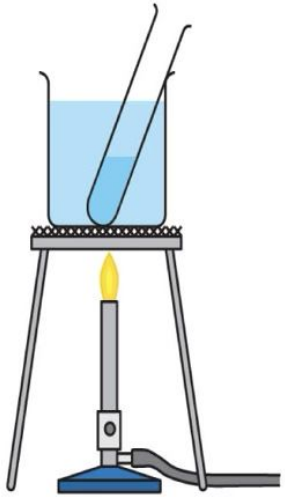
1. Add 2 cm^3 of Benedict's solution to 2 cm^3 of food sample in a test tube.
2. Shake the mixture.
3. Heat the contents in a boiling water bath for 2-3 minutes.

https://www.youtube.com/watch?v=Lt7RCIfudYQ

Benedict’s Test







A green, yellow or brick-red precipitate is formed if reducing sugar is present in trace, moderate or large amounts, respectively.

Results

Blue; Negative, no Trace of reducing sugar

Green; Positive, Trace amount of reducing sugar

Orange; Positive, Moderate amount of reducing sugar

Brick red; Positive, Large amount of reducing sugar

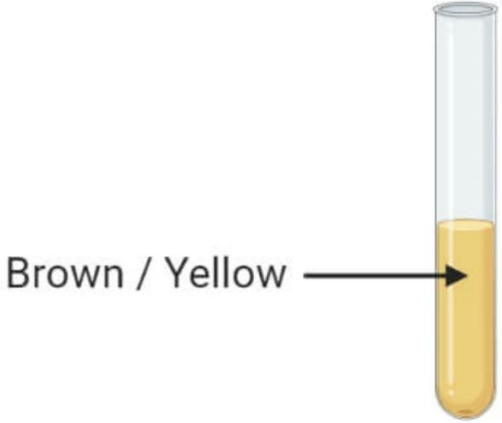
How can we test for starch?

By carrying out the **Iodine test**

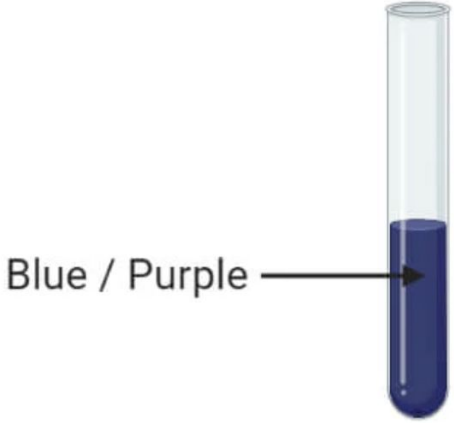
1. Add 2 cm^3 of food sample to a clean test tube.
2. Add a few drops of iodine solution to the test tube.

https://www.youtube.com/watch?v=J5eaJc854uQ

Results - A blue-black solution is formed if starch is present.



Negative Test (starch absent)



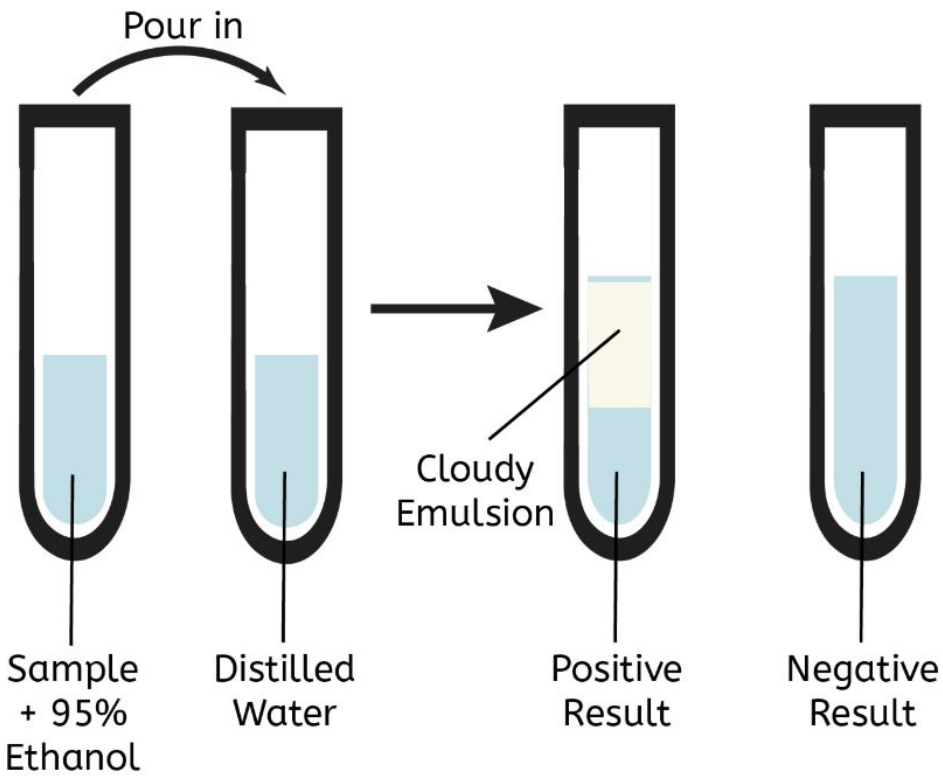
Positive Test (starch present)

How can we test for fats?

1. Add 2 cm^3 of ethanol to 2 cm^3 of food sample in a test tube.
2. Shake the contents of the tube vigorously.
3. Decant 3 cm^3 of water into the test tube and shake the mixture.

https://www.youtube.com/watch?v=_BcXylW7JFc

Results - A cloudy white emulsion is formed if fats are present.



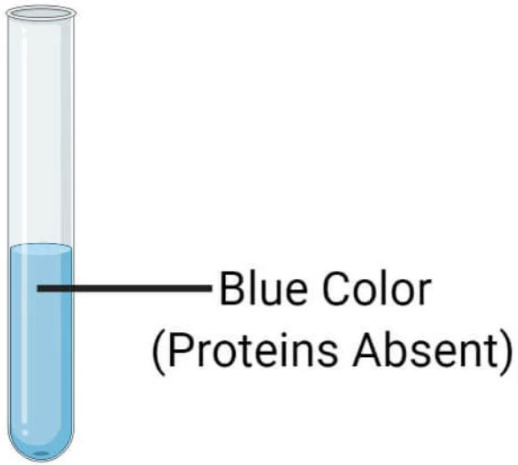
How can we test for proteins?

By carrying out the **Biuret test**

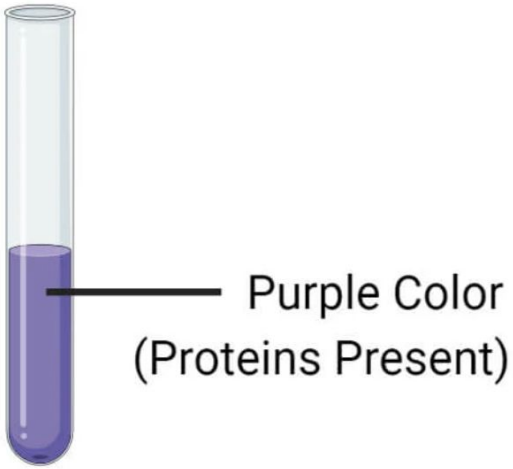
1. Add 2 cm^3 of sodium hydroxide solution to 2 cm^3 of food sample into a test tube.
2. Shake the mixture thoroughly.
3. Add 1% copper(II) sulfate solution drop by drop, shaking the mixture after each drop.

https://www.youtube.com/watch?v=ufec89A47uM

Results - A purple or violet solution is formed if proteins are present.



Negative Biuret Test



Positive Biuret Test

\- +

Benedict's test (reducing sugars)

blue → green → yellow → orange → brick red

Iodine test (starch)

brown orange → blue black

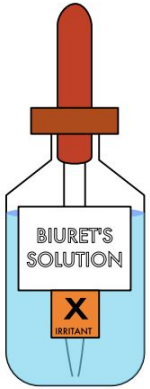
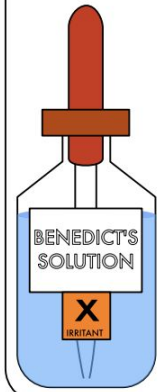
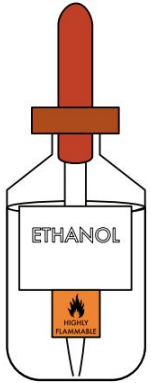
Biuret test (protein)

blue → purple

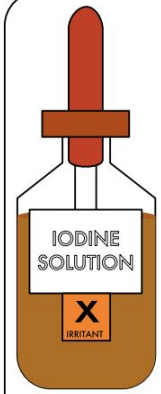
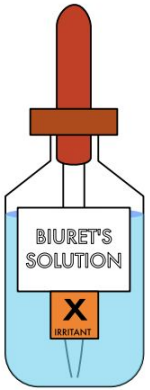
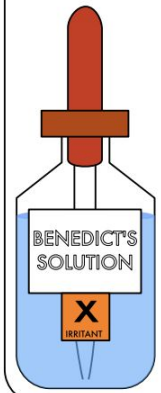
Emulsion test (lipid)

clear → milky emulsion

BIOCHEMICAL (FOOD) TESTS

CHEMICAL	TESTS FOR ...?	HOW TO CARRY OUT THE TEST	RESULT	CHEMICAL	TESTS FOR ...?	HOW TO CARRY OUT THE TEST	RESULT
							
							

BIOCHEMICAL (FOOD) TESTS

CHEMICAL	TESTS FOR ...?	HOW TO CARRY OUT THE TEST	RESULT	CHEMICAL	TESTS FOR ...?	HOW TO CARRY OUT THE TEST	RESULT
	Starch	1.) Add the iodine solution directly to the substance to be tested (in solid or liquid form) and look for a colour change.	Turns blue black with starch		Protein	1.) Add Biuret's to the solution/ suspension to be tested and look for a colour change.	Turns purple with protein
	Reducing Sugar	1.) Add Benedict's to the solution/ suspension to be tested. 2.) Heat for 2 mins in a water bath at boiling point and look for a colour change.	Turns brick red with reducing sugars (green/ yellow/ orange if less sugar present)		Lipid (known as the Emulsion test)	1.) Add ethanol to the solution/ suspension to be tested and shake thoroughly. 2.) Then add water and look for a colour change.	Turns cloudy/ milky with lipid

Energy in food

- Describe how to find out the energy content of a piece of food

Practical: Energy Content of a Food Sample

We can investigate the energy content of food in a simple calorimetry experiment

Apparatus

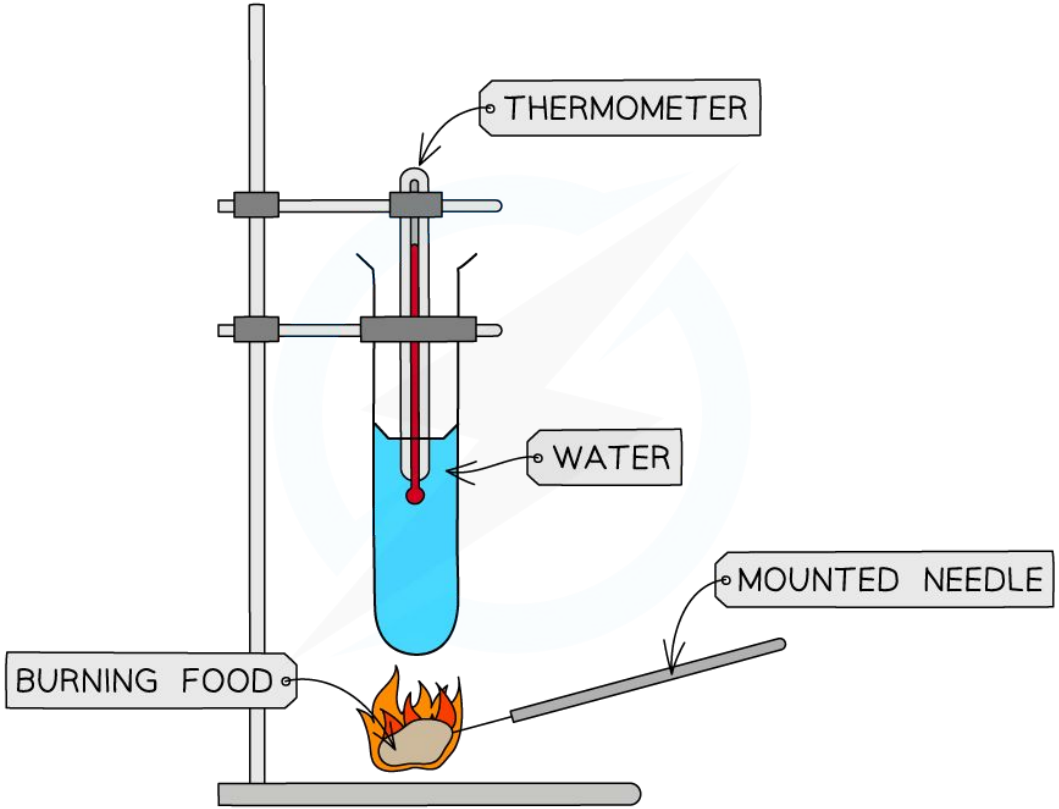
- Boiling tube
- Boiling tube holder
- Bunsen burner
- Mounted needle
- Measuring cylinder
- Balance/scales
- Thermometer
- Water
- Food samples

Method

- Use the measuring cylinder to measure out 25 cm3 of water and pour it into the boiling tube
- Record the starting temperature of the water using the thermometer
- Record the mass of the food sample
- Set fire to the sample of food using the bunsen burner and hold the sample 2 cm from the boiling tube until it has completely burned
- Record the final temperature of the water
- Repeat the process with different food samples
    - E.g. popcorn, nuts, crisps

Investigating the energy content of food samples diagram

Different food samples can be burned in a simple calorimetry experiment to compare the energy contents of the samples



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Results

- The larger the increase in water temperature, the more energy is stored in the sample
- We can calculate the energy in each food sample using the following equation:

$$\text{energy transferred per gram of food } (J) = \frac{\text{mass of water } (g) \times \text{ temperature increase } (°C) \times 4.2}{\text{mass of food sample } (g)}$$

- 4.2 kJ is the specific heat capacity of water, meaning that it is the energy required to raise 1 kg of water by 1 °C
- 1 cm^3 of water has a mass of 1 g

The energy content of food samples table

Food sample	Mass of water / g	Mass of food / g	Initial water temperature / °C	Final water temperature / °C	Change in water temperature / °C	Energy transferred per gram of food (J)
Popcorn	25	8.5	20.5	31.2	10.7	132.2
Walnut	25	8.1	20.4	34.1	13.7	177.6

Think-Pair-Share: Discuss Limitations

What are the limitations?

What can be done to reduce the limitations?

Limitations

- Incomplete burning of the food sample
  - Some food may not burn completely, so not all of the energy of the food is released
  - Solution: relight the food sample until it no longer lights up
- Heat energy loss to the surroundings
  - Some of the energy from the burning food escapes into the air instead of heating the water
  - This can make the calculated energy values lower than the true values
- Ways to reduce heat loss:
  - Simple improvements:
    - Keep the boiling tube close to the burning food (but safely)
    - Use a piece of aluminium foil around the tube to reduce heat loss
  - An alternative is to use a calorimeter:
    - A calorimeter is a piece of equipment that captures almost all of the heat released when food burns
    - Instead of burning food in open air to heat water in a boiling tube, the food is burned inside the calorimeter
    - This reduces heat lost to the surroundings and transfers more of the food's energy to the water. As a result, the calculated energy content (from measuring the temperature change in the water) is more accurate and closer to the true energy stored in the food

Proteins	Lipids
• Why do we need to eat protein – what role does it have in our body? • Name some foods rich in protein • What is the structure of a protein – what are they made of? • Name some examples of molecules/tissues made of protein in our body	• Why do we need to eat lipids – what roles do they have in our body? • Name some foods rich in lipids • What is the structure of a lipid – what are they made of? • EXTENSION: what is the difference between a fat and an oil?

Carbohydrates

- Why do we need to eat carbohydrates – what roles do they have in our body?
- Name some foods rich in carbohydrates
- What is the structure of starch? Sketch it.
- Name some sugars
- Name some polymers of glucose
- What is fibre and why is it needed in our diet

Nutrients

Carbohydrates

Organic compounds made up of carbon, hydrogen and oxygen in a ratio of 1:2:1

may be

- Monosaccharides (single sugars), e.g. glucose, fructose and galactose
- Disaccharides (double sugars), e.g. maltose, lactose and sucrose
- Polysaccharides (complex carbohydrates) made up of many monosaccharides

Identification
Test for reducing sugars: A reducing sugar gives a brick-red precipitate when boiled with Benedict's solution. Sucrose is a non-reducing sugar.

e.g.

- Starch → main form of storage for carbohydrates in green plants; and gives a blue-black colour with iodine (test for starch).
- Glycogen → main form of storage for carbohydrates in animals and fungi.
- Cellulose → forms the cell walls in plants; and cannot be digested by mammals, but is used as fibre in their diet.

Fats

Organic compounds made up of carbon, hydrogen and oxygen but they contain much less oxygen in proportion to hydrogen

Basic units
Fats are hydrolysed to form fatty acids and glycerol.

Identification
Test for fats: A cloudy white emulsion is formed in the ethanol emulsion test.

Proteins

Organic compounds made up of carbon, hydrogen, oxygen and nitrogen. Sulfur may also be present.

Basic units
Proteins consist of amino acids linked together by peptide bonds.

amino acids
↓
polypeptides
↓
proteins

Identification
Test for proteins: Proteins give a violet colouration with biuret solution.

Water

Inorganic nutrient. Water does not contain carbon.

Functions of water
In animals, water:

- is a medium for chemical reactions to occur;
- transports digested food products, excretory products, and hormones from one part of the body to another;
- is an essential part of protoplasm, lubricants, digestive juices and blood;
- is essential for hydrolysis; and
- helps to control body temperature.

In plants, water:

- is essential for photosynthesis;
- is needed to keep plant cells turgid;
- transports mineral salts from the roots to the leaves; and
- transports food substances from the leaves to other parts of the plants.